

常数项级数 + 正数项级数

$$1 (1) \sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$$

$$S_n = \sqrt{2} - 1 + \sqrt{3} - \sqrt{2} + \dots + \sqrt{n+1} - \sqrt{n} = \sqrt{n+1} - 1 \rightarrow \infty, n \rightarrow \infty$$

级数发数

$$(2) \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)}$$

$$S_n = \frac{1}{2} (1 - \frac{1}{3} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1})$$

$$= \frac{1}{2} (1 - \frac{1}{2n+1}) = \frac{n}{2n+1}$$

$$S_n \rightarrow \frac{1}{2}, n \rightarrow \infty \quad \text{级数收敛}$$

$$2 (1) -\frac{8}{q} + \frac{8^2}{q^2} - \dots + (-1)^n (\frac{8}{q})^n$$

$$= -\frac{8}{q} + (-\frac{8}{q})^2 + \dots + (-\frac{8}{q})^n$$

$\Rightarrow$ 为等比数列, $|q| \leq 1$, 收敛.

为级数收敛.

$$(2) \frac{1}{3} + \frac{1}{6} + \frac{1}{9} + \dots + \frac{1}{3n}$$

$$= \frac{1}{3} (1 + \frac{1}{2} + \dots + \frac{1}{n})$$

$\therefore 1 + \frac{1}{2} + \dots + \frac{1}{n}$ 发数. 利用性质 qS_n 与 S_n 相同

$$1 + \frac{1}{3} (1 + \frac{1}{2} + \dots + \frac{1}{n}) \text{ 也发数}$$

级数发数

$$(2) \frac{1}{3} + \frac{1}{13} + \frac{1}{23} + \dots + \frac{1}{n^3}$$

$$u_n = \frac{1}{n^3} = 3^{-\frac{1}{n}} \rightarrow 1, n \rightarrow \infty$$

$\therefore 1 \neq 0 \Rightarrow$ 级数发数

$$(4) (\frac{1}{2} + \frac{1}{3}) + (\frac{1}{2^2} + \frac{1}{3^2}) + \dots + (\frac{1}{2^n} + \frac{1}{3^n})$$

$$= \sum_{n=1}^{\infty} \frac{1}{2^n} + \sum_{n=1}^{\infty} \frac{1}{3^n}$$

$=$ 收敛 + 收敛 $=$ 收敛

$\Rightarrow$ 级数收敛

$$3 (1) 1 + \frac{1+2}{1+2} + \dots + \frac{1+n}{1+n^2} + \dots$$

$$u_n = \frac{1+n}{1+n^2} > \frac{1+n}{n+n^2} = \frac{1}{n}$$

$\sum_{n=1}^{\infty} \frac{1}{n}$ 发数, $\therefore$ 级数发数

$$(2) \frac{1}{n+b} + \frac{1}{2n+b} + \dots + \frac{1}{na+b}$$

$$u_n = \frac{1}{na+b} > \frac{1}{n} (\frac{1}{a+b})$$

$\therefore > \frac{1}{n} (1 + \frac{1}{2} + \dots + \frac{1}{n})$, 级数发数

级数收敛发数

$$(2) \frac{1}{2 \cdot 5} + \frac{1}{3 \cdot 6} + \dots + \frac{1}{(n+1)(n+4)}$$

$$u_n = \frac{1}{(n+1)(n+4)}$$

$$\lim_{n \rightarrow \infty} \frac{u_n}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 + 5n + 4} = 1$$

级数 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, $\therefore$ 级数收敛

$$(4) \sinh \frac{x}{2} + \sinh \frac{x}{2^2} + \dots + \sinh \frac{x}{2^n} + \dots$$

$$\lim_{n \rightarrow \infty} \frac{\sinh \frac{x}{2^n}}{\frac{1}{2^n}} = x, \sum_{n=1}^{\infty} \frac{1}{2^n} \text{ 收敛}$$

$\therefore$ 级数收敛

$$(5) \sum_{n=1}^{\infty} \frac{1}{1+a^n} \quad (a>0)$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{1+a^n}}{\frac{1}{1+a^{n+1}}} = \lim_{n \rightarrow \infty} \frac{a^n}{1+a^n} = \begin{cases} \frac{1}{2} & a=1 \\ 1 & a>1 \\ 0 & 0<a<1 \end{cases}$$

$\Rightarrow$ 当 $a>1$ 时, $\sum_{n=1}^{\infty} \frac{1}{1+a^n}$ 收敛
当 $0<a<1$ 时, $\sum_{n=1}^{\infty} \frac{1}{1+a^n}$ 发散

$$4(1) \frac{3}{1 \cdot 2} + \frac{5^2}{2 \cdot 2^2} + \dots + \frac{3^n}{n \cdot 2^n} + \dots$$

$$u_n = \frac{3^n}{n \cdot 2^n}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{3^{n+1}}{(n+1)2^{n+1}} \cdot \frac{n \cdot 2^n}{3^n} = \lim_{n \rightarrow \infty} \frac{3}{2} \cdot \frac{n}{n+1} = \frac{3}{2} > 1$$

级数发散

$$(2) \sum_{n=1}^{\infty} \frac{n^2}{3^n}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{(n+1)^2}{3n^2} = \frac{1}{3} < 1$$

级数收敛

$$(3) \frac{3}{2} + 2 \left(\frac{3}{2}\right)^2 + \dots + n \left(\frac{3}{2}\right)^n$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1) \left(\frac{3}{2}\right)^{n+1}}{n \left(\frac{3}{2}\right)^n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{3}{2} = \frac{3}{2} < 1$$

级数收敛

$$(4) \frac{1^4}{1!} + \frac{2^4}{2!} + \dots + \frac{n^4}{n!} + \dots$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^4}{(n+1)!} \cdot \frac{n!}{n^4} = \frac{(n+1)^4}{(n+1)n^4} = \lim_{n \rightarrow \infty} \frac{1}{n} \left(1 + \frac{1}{n}\right)^4 = 0 < 1$$

级数收敛

$$(5) \sum_{n=1}^{\infty} \frac{n+n^2}{2^{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1) + (n+1)^2}{2^{n+2}} \cdot \frac{2^{n+1}}{n+n^2} = \lim_{n \rightarrow \infty} \frac{n+1}{n} \times \frac{1}{2} = \frac{1}{2} < 1$$

级数收敛

$$5(1) \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} < 1$$

$\therefore$ 级数收敛

$$(2) \sum_{n=1}^{\infty} \frac{1}{[\ln(n+1)]^n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{1}{\ln(n+1)} = 0 < 1$$

级数收敛

$$(3) \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n = \left[\left(1 - \frac{1}{n}\right)^{-n}\right]^{-1} = \frac{1}{e} < 1$$

级数收敛

$$(4) \sum_{n=1}^{\infty} \left(\frac{b}{a^n}\right)^n, \quad a \rightarrow a, \quad (a \neq b)$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \frac{b}{a^n} = \frac{b}{a}$$

当 $b < a$ 时, 级数收敛

$b > a$ 时级数发散

$$(5) \sum_{n=1}^{\infty} \left(\ln \frac{1}{n}\right)^{n^3}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \left(\ln \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2n^2}\right)^{-2n^2} = e^{-\frac{1}{2}} < 1$$

级数收敛

六. 证明.

$$(1) \lim_{n \rightarrow \infty} \frac{n!}{2 \cdot 5 \cdot 8 \cdots (3n-1)} = 0$$

$$u_n = \frac{n!}{2 \cdot 5 \cdot 8 \cdots (3n-1)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!}{2 \cdot 5 \cdot 8 \cdots (3n-1)(3n+2)} \cdot \frac{2 \cdot 5 \cdot 8 \cdots (3n-1)}{n!} = \frac{1}{3} < 1$$

$$\sum_{n=1}^{\infty} u_n \text{ 收敛}$$

$$\therefore \text{收敛} \Rightarrow \lim_{n \rightarrow \infty} \frac{n!}{2 \cdot 5 \cdot 8 \cdots (3n-1)} = 0$$

$$(2) \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{\frac{(n+1)!}{(n+1)^{n+1}}}{\frac{n!}{n^n}} = \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{n+1}{n}\right)^n} = \frac{1}{e} < 1$$

$$\sum_{n=1}^{\infty} \frac{n!}{n^n} \text{ 收敛} \Rightarrow \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

7. 设 $u_n > 0 (n=1, 2, \dots)$, 证明: 如果级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 则级数 $\sum_{n=1}^{\infty} u_n^2 \Rightarrow \sum_{n=1}^{\infty} \frac{\sqrt{u_n}}{n}$ 都收敛.

解: 先证明 $\sum_{n=1}^{\infty} u_n^2$ 收敛

$$\because \sum_{n=1}^{\infty} u_n \text{ 收敛}, \Rightarrow \lim_{n \rightarrow \infty} u_n = 0, \quad n \rightarrow \infty \text{ 时}, u_n < 1, \quad u_n^2 \leq u_n$$

$$\therefore \sum_{n=1}^{\infty} u_n^2 \text{ 也收敛}$$

再证明 $\sum_{n=1}^{\infty} \frac{\sqrt{u_n}}{n}$ 收敛: $\frac{\sqrt{u_n}}{n} \leq \frac{1}{2} \left(u_n + \frac{1}{n^2} \right)$ (基本不等式)

$$\therefore \sum_{n=1}^{\infty} u_n \geq \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ 均收敛}$$

$$\sum_{n=1}^{\infty} \left(u_n + \frac{1}{n^2} \right) \text{ 收敛}, \quad \sum_{n=1}^{\infty} \left[\frac{1}{2} \left(u_n + \frac{1}{n^2} \right) \right] \text{ 收敛}$$

$$\therefore \sum_{n=1}^{\infty} \frac{\sqrt{u_n}}{n} \text{ 收敛}$$